induced by the pituitary hormones (LH and FSH) and ovarian hormones (estrogen and progesterone).

Info bit

Menstruation is a periodical phenomenon that continues from puberty to menopause. This will happen if the released ovum is not fertilized by the sperm. Lack of menstruation generally indicates pregnancy.

17.10 Fertilization to Foetal Development

Fertilization

Fertilization in human is internal and occurs in the oviduct of the female genital tract. It takes place usually in the ampulla of the fallopian tube. An oocyte is alive for about 24 hours after it is released from the follicle. Fertilisation must take place within 24 hours. The sperm enters into the ovum and fuses with it, resulting in the formation of a '**zygote**'. This process is called fertilization. The zygote is a **fertilized ovum**.

Cleavage and Formation of Blastula

The first cleavage takes place about 30 hours after fertilization. Cleavage is a series of rapid mitotic divisions of the zygote to form many celled blastula (**Blastocyst**) which comprises an outer layer of smaller cells and inner mass of larger cells.

Implantation

The blastocyst (fertilized egg) reaches the uterus and gets implanted in the uterus. The process of attachment of the blastocyst to the uterine wall (**endometrium**) is called implantation. The fertilized egg becomes implanted in about 6 to 7 days after fertilization.

Gastrulation

The transformation of blastula into gastrula and the formation of **primary germ layers** (ectoderm, mesoderm and endoderm) by rearrangement of the cells is called gastrulation. This takes place after the process of implantation.

Info bits

Normally one egg matures in the ovary each month. Ovulation is the rupture of the follicle releasing the egg or ovum . The uterus prepares itself to receive the fertilized egg every month. The uterine lining becomes thick and spongy for implantation of the fertilized egg.

Events leading to when fertilization occurs and does not occur

If fertilization takes place the corpus luteum persists, continues to secrete progesterone maintains the thickened state of uterine wall and prevents maturation of another follicle till the end of pregnancy

If fertilization does not occur, corpus luteum degenerates, the egg disintegrates and the uterine lining slowly breaks, discharged as blood and mucus leading to menstrual events.

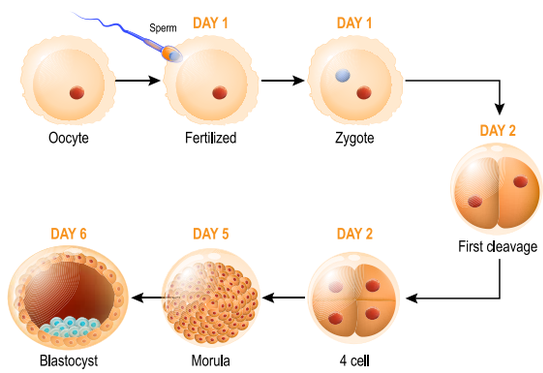


Figure 17.18 Developmental stages of zygote from cleavage to blastocyst formation

Organogenesis

The establishment of the germ layers namely ectoderm, mesoderm and endoderm initiates the final phase of embryonic

10th_Science Unit-17.indd 252 06-12-2021 18:43:16